FARIA · 7 NOVEMBER 2026 · A LESSON FOR JEFF FRANZEN

Jeremiah Dixon: the surveyor in the partnership

Jeremiah Dixon was born in 1733 at Cockfield, in County Durham, into a Quaker family with means. He trained as a surveyor and worked as an amateur astronomer. When the Royal Society needed a second man for the 1761 transit, John Bird's recommendation is the account to trust, and to trust only as far as it goes. Bird made instruments and knew both worlds, the workshop and the observatory. A family story about an examination at Woolwich is thinner: the Ordnance minutes that would confirm it have not been produced. Say "recommended through Bird's network," not "we watched them meet."

A Quaker surveyor in the north of England already knew how a chain lies, how a slope cheats a distance, and how a neighbor disputes a hedge. That craft is not astronomy, and astronomy is not that craft. The Royal Society needed both on one road, which is why the pairing matters more than which man stood an inch to the left.

Dixon was not a spare hand hired to hold Mason's chain. At the Cape he observed. On the boundary, from 1763, surveying was his profession and astronomy was Mason's daily habit. The work needed both. A chain of standard length, a slope corrected, a trial line checked by a second measurement: those are the surveyor's corrections, and they are legitimate. They are not a confession that the first number was a scandal.

In October 1767 the Six Nations escort stopped at the war path. Their leader told the party that this was as far as his commission from the chiefs ran, and that he would not go one step farther west. That sentence is about a limit set by the people who sent him. It is not a license to invent a speech about fear, and it is not a permission slip written by the geometry. Some of the hired men, separately, had already refused to cross a river for fear of the Shawnee and the Delaware. Two facts. Do not melt them into one slogan.

Dixon died in Britain in 1779. The line he helped lay is the next lesson. Mason's lesson has the Greenwich years and the rule that a place must be known before a sight is interpreted. The pair only works if you can see both men.

Norway in 1769: Dixon and Chappe compared

Who observed the transit of 3 June 1769 better, Jeremiah Dixon in Norway or Jean-Baptiste Chappe d'Auteroche in Baja California? Weather, instruments, timing, and later praise all belong in the question. They do not add up to a winner.

Dixon was not at Vardo with Maximilian Hell, and he was not beside William Bayly. Nils Voje Johansen's account puts Bayly at Honningsvag, near the North Cape, and Dixon at Rypeklubben, just outside Hammerfest. Cloud held Dixon until 20 May, his first clear observations. On 3 June the weather was cloud and gales. Johansen writes that both men "failed to observe the transit satisfactorily." That sentence is about the transit. Dixon and Bayly still published the positions of the observatories, altitudes of the Sun, meridian transits, the solar eclipse that followed, and notes on the weather. A missed contact is not an empty expedition.

Dixon's own instrument list is not known. Johansen reconstructs Bayly's from the papers they printed: a quadrant of one-foot radius by Bird, a two-foot reflector by Dollond, a four-foot transit instrument by Bird with a Dollond lens, and clocks by Shelton, including an astronomical clock with a gridiron pendulum. Bayly said Dixon's instruments were similar, except that Dixon had no dipping needle. That is a reconstruction, not a case opened and listed. A contact time to the second, for Dixon, was not established here. Bayly wrote down a hazy contact of his own. Those seconds stay his.

Chappe's list is his own, quoted by Espinoza Meléndez: a three-foot-radius quadrant by Canivet, a smaller English quadrant of one-and-a-half-foot radius, a transit instrument, a parallactic machine, Dollond telescopes of ten feet and three feet, and a pendulum clock by Berthoud. He had checked the clock and the telescopes, with repeated observations of the Sun, before 3 June. The landing, the illness, and the death on 1 August are on the Venus lesson. So is the longitude. Cassini's adopted figure in the 1772 volume comes from the transit, not from an improvement measured by the eclipse of 18 June. A seconds-level accuracy for Chappe himself was not established. Cassini's 20 or 30 seconds of time is a caution about the Jupiter-moon method, not a personal error bar.

Later pages praise the Baja work. A Smithsonian exhibition describes the conditions as ideal. An American Scientist article calls the observations among the best made of either transit, and the position work unusually accurate for hard fieldwork. That is later praise, from those pages, not a jury sitting in 1769. No formal ranking of Dixon against Chappe was found, and nothing found here says Dixon was the less skillful man. The sky at Hammerfest failed him. No weighting by Encke that picks a winner was found.

The illustrated lesson is download the PDF.

Sources

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Key facts

- Dixon was born in 1733 at Cockfield, County Durham, and worked as a surveyor and amateur astronomer.
- Bird's recommendation brought him to Mason. The literal first meeting is unknown, and the Woolwich exam story is unverified.
- He observed at the Cape in 1761 and surveyed the boundary from 1763. Planned corrections are part of the method.
- In October 1767 the escort stopped at the commission's geographic limit. Dixon died in Britain in 1779.
- In 1769 Dixon was at Rypeklubben, near Hammerfest, and Bayly at Honningsvag. Hell at Vardo was a separate station. Cloud until 20 May, and cloud and gales on 3 June. Johansen's words are that both "failed to observe the transit satisfactorily." They still published the place and other observations.
- Dixon's exact instruments are not known. Johansen's list reconstructs Bayly's kit, which Bayly called similar except for the dipping needle. Chappe's own list names the Canivet and English

quadrants, a transit instrument, a parallactic machine, two Dollond telescopes, and a Berthoud clock.

- Later histories praise Chappe's Baja observations. No contemporary ranking, and no Encke weighting, was found. The adopted longitude in the 1772 volume is the transit's, not an eclipse improvement. The weather, not a judgment of skill, is what failed Dixon.